

Ibrutinib in B-Cell Malignancies: Mechanism of Action, Clinical Applications, Safety, and Resistance

Oncology Interest Group

1. Introduction

The treatment of cancer has been changed and developed along the history of medicine due to the research and trials that have been conducted. It started with aggressive and dangerous treatments like chemotherapy and radiation, but over time with conduction of much research, and an increase in knowledge, a new type of therapy has appeared in the field. The researchers found that we can develop drugs that act specifically on specific molecules on cancer cells exclusively, which is a huge step in the future of cancer therapy. The drugs that act through this mechanism are called targeted therapy drugs [1]. Bruton Kinase inhibitors (BTKi) such as ibrutinib represent an important class of targeted therapies. This article discusses ibrutinib's pharmacological features, mechanisms of action, current clinical applications, adverse effects, and resistance.

2. BTK Signaling Pathway

Bruton's tyrosine kinase (BTK) is a non-receptor tyrosine kinase, that has a critical role in B-cell survival, activation, and clonal expansion as a downstream catalytic agent, propagating signals initiated by activated B-cell receptors (BCRs), and as a non-catalytic scaffolding protein, aiding in the assembly of toll-like receptors, adhesion molecules, and other kinases [3].

Amplified BTK activity is observed in certain B-cell malignancies, such as Mantle cell, Waldenström Macroglobulinemia, follicular lymphoma, and diffuse large B-cell lymphoma. This has made BTK a therapeutic target for inhibitory agents used in treating these cancers [3].

BTK is a protein made from 659 amino acids and contains five signaling domains [8]. On the N-terminus, the pleckstrin homology (PH) aids in its membrane attachment. The Tec homology (TH), which is proline-rich, mediates interactions with other proteins. The Src homology 3 (SH3) facilitates interactions with proline-rich sequences, and Src homology 2 (SH2) serves as a substrate recognition site. BTK's catalytic action is mediated by the kinase domain on its C-terminus [3].

BCR signaling is mediated by a downstream cascade of secondary messengers, starting with Lyn tyrosine kinase, which directly interacts with the BCR complex. Next, spleen tyrosine kinase transduces the signal from Lyn to BTK via PI3K and MAP kinases [3].

After BTK activation, it is recruited to the plasma membrane, where it gets phosphorylated and acts on various substrates, most importantly phospholipase C γ 2 (PLC γ 2), leading to its activation [3].

PLC γ 2 cleaves PIP2 into IP3 and DAG. IP3 releases intracellular Ca⁺⁺ stores, and DAG activates protein kinase C. This is followed by the activation of the RAS/ERKs pathway and the IKK (inhibitor of the nuclear factor- κ B kinase) complex. These signaling events ultimately result in the activation of transcription factors such as NF- κ B, MYC, and ELK, promoting transcription of genes involved in survival and proliferation [3].

The integrity of these pathways maintains the intensity and duration of BCR signaling in response to antigenic stimulation [3].

3.Bruton Kinase Inhibitors

BTK inhibitors (BTKi) are a group of drugs that have been approved in cancer therapy because of their action in blocking and suppressing the BTK enzyme function in B-cells. They are used mainly in blood cancers like mantle cell lymphoma (MCL), chronic lymphocytic leukemia (CLL), and small lymphocytic lymphoma (SLL) [2].

Although it was found that this group has a good role in inhibiting tumor cell movement to the blood circulation, which makes the tumor easier to deal with, it is also used in many solid tumors alongside chemotherapy, like prostate, ovarian, and brain cancers [2].

Additionally, they can be used in the treatment of autoimmune diseases like MS, SLE, and RA, which are caused by the production of autoantibodies due to their effect on B-cells. They are used with other immune-modulator drugs and can replace B-cell depleting agents due to their specificity [2].

4, Mechanism of action and pharmacology

Ibrutinib binds covalently (irreversibly) to the active site of BTK, with a cellular IC₅₀ of 11 nM, effectively blocking BTK-mediated signaling in a dose-dependent manner. Thereby, it reduces survival signaling, leading to reduced proliferation and induced apoptosis in malignant cells, even in the presence of prosurvival factors such as CD40L, B-cell activating factor, and protective stroma [4].

Additionally, it compromises migration by impairing cellular responses towards stromal (homing) chemokines (CXCL12 and CXCL13) and by downregulating the production of BCR-dependent chemokines secreted by CLL cells (CCL3 and CCL4), leading to egression of lymphocytes from their protective stroma towards peripheral blood [4].

This “lymphocyte redistribution” phenomenon is responsible for the transient lymphocytosis that occurs in the first few weeks of therapy, and it is subsequently followed by gradual lymphocyte apoptosis. While the rise of lymphocyte count resembles worsening disease, it is NOT an indicator of disease progression or failure of treatment, as other disease burden markers, such as lymph node size, continue to show improvement. Resolution of LR occurs within 4 weeks, although it may persist for months and is thought to be accelerated in individuals with unmutated IGHV. However, the mechanisms behind that association are not known [4].

Ibrutinib has a rapid absorption rate through the oral route. It peaks in concentration 1-2 hours after ingestion, and exhibits a half-life of 2-3 hours. Because it can bind covalently to BTK for at least 24 hours, continuous inhibition can be achieved with a single daily dose despite its short half-life. [4] The approved daily dose is 560 mg for mantle cell lymphoma patients and 420 mg for patients with CLL and WM [9].

The drug is metabolized in the liver; thus, hepatic impairment would significantly increase plasma drug concentrations and, therefore, requires greater caution. CYP3A inducers or inhibitors would also substantially alter the drug level; concurrent administration should be avoided [4].

5. Clinical applications

BTK inhibitors are currently used in the treatment of some NHLs, most notably Chronic Lymphocytic Leukemia (CLL), Mantle Cell Lymphoma (MCL), and Waldenström Macroglobulinemia (WM) [3].

The innovation of BTK inhibitors has been revolutionary for NHL therapy, given BCR’s major role in the pathogenesis of CLL, MCL, and WM.

Ibrutinib is a first-generation BCR antagonist, that works by covalently binding to cysteine (Cys)-481 in the active site of BTK, irreversibly inhibiting its tyrosine kinase activity at the site tyrosine (Tyr)-223 [3].

Ibrutinib’s ability to induce apoptosis (via caspase activation) and inhibit cell proliferation in CLL cells was effectively demonstrated in vitro experiments, which encouraged the commencement of clinical trials [7].

The earliest clinical experiments included CLL and MCL patients, and later encompassed challenging NHL cases, including refractory and relapsed patients[3] .

Ibrutinib got the FDA approval in 2013 initially for the treatment of relapsed MCL [7]and later on, for CLL and WM, due to the optimistic results provided by clinical trials. [3].

6. Adverse effects

BTK inhibitors are administered continuously over a prolonged period until disease progression diminishes or major toxicities occur. **Common adverse effects of ibrutinib include:** Hemorrhage, infections, cytopenias, cardiac arrhythmias, hypertension, second primary malignancies, tumor lysis syndrome, embryo-fetal toxicity. 2 of the side effects will be discussed in detail [5].

1. Bleeding and Haemorrhage

A common ADR, caused by both on-target and off-target mechanisms. BTK directly plays a role in normal platelet function, as in signaling, adhesion, and aggregation; thereby, BTK inhibition impairs regular platelet function and increases the risk of bleeding. Off-target interactions with TEC kinases are also implicated in the susceptibility to bleeding [5].

Coadministration of antithrombotic agents such as Warfarin alongside Ibrutinib further amplifies the risk of hemorrhage, specifically subdural hematomas, and should therefore be avoided. This raises therapeutic challenges in preventing blood clots associated with atrial fibrillation, which can occur as an adverse reaction to Ibrutinib [5].

Ibrutinib should be withheld 3 to 7 days before and after surgery, as with any blood-thinning agent, as a precaution against hemorrhage [5].

2. Infectious Complications

While most notably seen with Ibrutinib, all BTK blockers are associated with immunosuppressor effects that may lead to opportunistic infections or reactivation of dormant infections [5]. Ibrutinib exerts an inhibitory effect on various immune cells, including T-cells (through off-target interactions with IL-2-inducible T-cell kinase), Natural killer cells (Inhibition of antibody dependent cellular cytotoxicity), and macrophages (impaired phagocytosis) [5].

Co-administration of other immunomodulators, such as corticosteroids or immunosuppressants, increases the risk of infections [5].

7. Resistance Mechanism

BTK mutations (C481S)

During ibrutinib therapy, some patients do not respond to the drug as well after a time from starting the treatment. This is caused by genetic mutations that occur in the targeted BTK gene on B cells. The most common mutation is C481S, which is specifically for the older covalent generations of BTKi like ibrutinib. This genetic change is characterized by the substitution of cysteine by serine at position 481 on the protein, and this alteration prevents the binding of covalent BTKi to the sulfhydryl (SH) on the cysteine residue, which is the actual site of binding [6].

PLC γ 2 mutation

PLC γ 2 is a protein located downstream of BTK in the signaling cascade of B-cells. Normally, PLC γ 2 is activated by BTK through phosphorylation mechanisms during the signaling pathway. When a mutation occurs in the PLCG2 gene, it is altered in a way that it doesn't need to be activated by BTK (phosphorylation-independent). As a result, the proliferation pathway of malignant B-cells will continue; the usage of BTKi with this mutation becomes less effective on the malignant cells. The types of this mutation are R665W, L845F, and S707Y [6].

8. Future perspectives

During the use of BTKi, many problems appeared in the field that need to be taken into consideration. The first-generation inhibitors, like ibrutinib, have lower specificity towards BTK compared to the 2nd generation and can interact with other kinases such as epidermal growth factor receptor(EGFR), which manifests as unwanted adverse effects [2].

Another problem with the 1st generation is its metabolism with CYP3A4 liver enzymes, which leads to interactions with other drugs that may decrease their efficacy or increase their toxicity[2].

To overcome these problems, new generations of BTKi have been developed. Second-generation BTK inhibitors like Zanubrutinib and Acalabrutinib have more specificity and fewer drug-drug interactions and are covalent (irreversible) BTK inhibitors [2].

Third-generation BTK inhibitors such as Pirtobrutinib are non-covalent (reversible) BTK inhibitors [2].

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